ANDREW NEIL

AI ENGINEERING | AGENTIC SYSTEMS | PYTHON BACKEND | APPLIED ML

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Portfolio | LinkedIn | GitHub

PROFILE

Computer Engineering student graduating in 2027, building practical AI agents and backend systems in Python. Experience spans LLM orchestration, tool execution and reliability testing; quantitative data pipelines and historical strategy evaluation; computer vision; and IoT-enabled financial workflows. Brings a test-first approach, paid freelance delivery experience, a current Django backend internship, a competition-winning vision project, and three co-authored IJIRT papers.

PROJECT HIGHLIGHTS 336 passing OSCAR regression tests • 3 Upwork contracts / 2 five-star reviews • Competition-winning YOLOv8/OCR system • 3 IJIRT papers

TECHNICAL SKILLS

Programming & data: Python, JavaScript (React/Vite projects), Pandas, NumPy, Parquet, pandas-ta

Agentic AI & LLMs: LLM task decomposition, tool calling, multi-step workflows, structured JSON actions, workflow state, Gemini/OpenAI/Anthropic APIs, LangChain, model-provider routing with OmniRoute, local-model integration, token/cost tracking

Reliability & testing: pytest, regression testing, failure-mode analysis, payload validation, fresh-observation loops, outcome verification, bounded retries, mocked API/OS tests

Backend & product engineering: Django, FastAPI, REST APIs, React, Vite, Tailwind CSS, Flutter/Dart, Git/GitHub, Linux/CLI, SMTP notifications

Automation, vision & applied ML: Playwright, Selenium, PyAutoGUI, YOLOv8, OpenCV, OCR/ANPR, scikit-learn, PyTorch (earlier LSTM experimentation), backtesting, walk-forward evaluation

IoT & security: ESP32-S3, NFC/RFID, Bluetooth LE, JWT, document encryption, blockchain/smart-contract prototypes

Developing knowledge: RAG fundamentals: document chunking, embeddings and vector search (currently learning)

EXPERIENCE

Backend Engineer Intern | 4iStudio | *Aug 2026–Present*

- Contribute Django backend workflows to a polytechnic ERP supporting student records, attendance, marks entry, marksheets and certificate generation across institutional roles.

Freelance Python Developer | Upwork | *2023–2025 · 3 contracts*

- Built a LangChain CLI integrating OpenAI and Claude APIs to generate three response variants, support regeneration and user selection, and reuse the selected output as context.
- Delivered a website-availability monitor with alerts through a client-provided SMTP server, plus Selenium-based browser automation and scraping. Completed three contracts for international clients and earned two five-star reviews.

PROJECT PORTFOLIO

OSCAR / OSCAR 2.0 | GitHub repo | *Multi-agent AI desktop automation assistant | OSCAR 2.0 in active development*

Tech: Python, Flutter/Dart, Gemini, OpenAI/Anthropic APIs, OmniRoute, local-model integration, Playwright, PyAutoGUI, OpenCV/OCR, pytest

- Built a Windows-first assistant that converts natural-language requests into step-based workflows across browser, desktop UI and terminal agents, coordinated by a Python orchestration backend and Flutter desktop interface.
- Implemented screenshot-driven planning and action loops: the model proposes structured JSON actions, Python dispatches mouse/keyboard/scroll inputs, and fresh observations inform the next step. Added record/replay and persistent workflow state.
- Designed for failure: malformed tool payloads are validated before execution; follow-up observations reduce stale-state actions; outcome checks, bounded retries, duplicate-typing prevention and a simple-task fast path help avoid loops and unnecessary LLM calls.
- 336 automated regression tests pass. The suite uses mocked Windows/API paths and does not establish real-world click accuracy. OSCAR 2.0 work focuses on precision, completion speed, token efficiency, multi-model routing and token-usage tracking.

TradeAssist | *Quantitative trading research, model experimentation & decision support*

Tech: Python, Angel One SmartAPI, Pandas, NumPy, pandas-ta, Parquet, PyTorch (prior LSTM experiments)

- Built a live market-data ingestion and processing workflow and a historical Indian-equity backtesting engine with technical signals, trade simulation, position sizing, entry/exit rules, modeled transaction costs, 5-bps slippage, stop/trailing-stop logic, buy-and-hold comparisons and portfolio metrics.
- Compared strategy variants over 13 walk-forward folds, including buy-and-hold comparisons. Best documented IRFC SMA 20/50 crossover + 5% stop-loss result: +24.17% total return, 6.75% CAGR and −11.03% maximum drawdown across 86 trades (Jan 2023–Sep 2026). The recorded run included INR 73,390.71 in modeled transaction costs. Earlier iterations explored LSTM time-series forecasting; these metrics belong to the SMA strategy, not the LSTM experiment.
- Historical backtest, not live performance. The model uses assumed 5-bps slippage and modeled transaction costs; backtest results do not guarantee future performance.

PROJECT PORTFOLIO

FinVerse | *IoT-enabled B2B payment platform | Final-year major project in development*
Official project title: *FINVERSE: AN AI-POWERED BLOCKCHAIN FINANCIAL OPERATING SYSTEM WITH IOT AUTHENTICATION, CYBERSECURITY AND INTELLIGENT FINANCIAL AUTOMATION*
Tech: Django, Python, React 19/Vite/Tailwind CSS v4, Flutter, blockchain, ESP32-S3, NFC/RFID

- Developing FinVerse 2.0 as a B2B payment and machine-to-machine (M2M) commerce platform, combining a financial copilot, AI-assisted fraud detection, blockchain-oriented financial workflows and hardware-backed authentication.
- The prototype spans a React dashboard, Django backend, Flutter client and an ESP32-S3 secure NFC terminal. Hardware-backed authentication and integration are being developed alongside the financial workflow software.

Personal Portfolio & Local AI Assistant | Live portfolio | *Deployed portfolio and browser-based model demo*
Tech: React 19, Vite, Tailwind CSS v4, WebLLM, Qwen2.5-0.5B-Instruct (Q4f16 MLC)

- Built and deployed a personal portfolio at andrewneil.dev with a recruiter-focused landing experience, project links and a browser-based AI assistant.
- Integrated WebLLM with Qwen2.5-0.5B-Instruct (Q4f16 MLC) to demonstrate local, in-browser model inference rather than relying entirely on a remote chatbot API.

Dynamic Toll Charging System | IJIRT paper | *Competition-winning computer-vision and toll/wallet prototype*
Tech: Python, YOLOv8, OpenCV, OCR/ANPR, RFID/FASTag simulation

- Combined YOLOv8 vehicle detection, number-plate recognition and RFID-based vehicle identification in a simulated entry/exit, distance-based toll and wallet workflow.
- Correlated entry/exit vehicle records to simulate distance-based charges and a dummy-wallet update, combining plate recognition and RFID instead of relying on a single identifier. Won the project competition and led to a co-authored IJIRT paper.

KYC Smart Card | IJIRT paper | *RFID-based digital document and identity-verification prototype*
Tech: Django, Flutter, ESP32, RFID, Bluetooth LE, JWT, document/key encryption

- Co-developed a DigiLocker-inspired document platform with RFID smartcard verification and ESP32/Flutter Bluetooth authentication; contributed to Django backend workflows and hardware communication.
- The dual-mode design includes JWT-based sessions and encrypted document-management workflows. The 2025 paper covers smartcard/mobile authentication; the earlier 2024 paper documents the RFID-linked repository concept. The project combines software authentication flows with physical-card and mobile-device interactions.

RentEscrow DApp | *Smart-contract prototype*
Tech: Solidity, Hardhat

- Developed a RentEscrow smart-contract prototype with Hardhat and validated the contract workflow with six passing tests. Deployed to local development chain ID 31337; this was a local-chain test deployment, not a mainnet launch.

PUBLICATIONS

Dynamic Toll Charging System — IJIRT, Vol. 12, Issue 8 (2026), pp. 5716–5719.
KYC Smart Card (dual-mode authentication) — IJIRT, Vol. 12, Issue 1 (2025), pp. 646–652.
KYC Smart Card (RFID document repository) — IJIRT, Vol. 11, Issue 7 (2024), pp. 26–30.

EDUCATION

B.E. Computer Engineering | Xavier Institute of Engineering (XIE), Mumbai | *Expected 2027*
CGPA: 7.48/10 through Semester VI | Relevant coursework: Data Structures, DBMS, Operating Systems, Computer Networks, AI/ML and IoT.